APOORV KUMAR SINGH

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EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY, BHOPAL **2023 – 2027**

*B.Tech - Computer Science and Engineering (Cloud Computing) - **CGPA - 8.34*** ***Bhopal, Madhya Pradesh***

St Atulanand Residential Academy **2019 – 2022**

*Higher Secondary: **77.7%** (2020–2022) | Matriculation: **77.2%** (2019-2020)* ***Varanasi, Uttar Pradesh***

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL, Java, HTML, CSS, JavaScript
- **Tools & Technologies:** OpenCV, Git, GitHub, VS Code

PROJECTS

NewsVITa – Centralized College News & Events Portal | Web Development **[GitHub]** **Sep 2024 – Dec 2024**

- Developed a free, centralized web platform for campus news, events, and announcements, improving college-wide communication.
- Built the front-end using HTML, CSS, and JavaScript with responsive layouts and clean UI for enhanced usability.
- Designed an intuitive event listing and registration system for easy student access to campus updates.

CloudCart – Cloud-Native E-Commerce Platform | Full-Stack Development **[GitHub]** **Feb 2025 – Present**

- Engineered a scalable cloud-native e-commerce system featuring decoupled Spring Boot REST API backend and Angular standalone frontend.
- Implemented Spring Security with stateless JWT token authentication, role-based authorization (Admin/Seller/User), and JPA pagination.
- Built Amazon-style responsive UI with product filtering, shopping cart, persistent wishlist, and cloud deployment resilience.

Vplay – Real-Time Tennis Analysis System | Machine Learning **[GitHub]** **Jan 2025 – April 2025**

- Developed a real-time tennis analytics system using YOLO and PyTorch to detect and track players, rackets, and tennis balls from match videos.
- Implemented CNN-based models for ball trajectory prediction and bounce detection to enhance match analysis accuracy.
- Processed video frames using OpenCV for court detection, event labeling, and visual feedback.
- Designed a lightweight and scalable solution suitable for coaching, training, and professional match analysis.

EXTRACURRICULAR ACTIVITIES

- **Co-Lead, Team Management – AdVITya'26 Annual Sports Fest:** Coordinated with external college and university teams for participation and event-related communication. Managed participating teams and coordinated with the organizing committee for smooth event execution.
- Active participant in hackathons (Innovator badge at Accenture Innovation Challenge) and coding challenges.
- One of the Best maintainer in AcWoc'25 Hackathon.

CERTIFICATIONS

– MATLAB Onramp – MathWorks **15 Dec 2023**

– Prompt Design in Vertex AI – Google Cloud **31 Aug 2025**

– The Bits and Bytes of Computer Networking – Coursera **1 Dec 2025**